What wine-producing countries offer the best balance between price and quality?

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1 Introduction and context

Wine ranges from a cheap everyday household staple to a luxury product, and people typically expect expensive wine to be of better quality. Nevertheless, what truly interests consumers is where to get the best quality for their money. This project therefore asks: Which countries produce wines that are rated higher than their price predicts?

The project originally planned to measure the value with a points-to-price ratio. However, this technique would mostly reward cheap wines, as the points given in the wine reviews only range from 80 to 100, while the prices vary by a factor of almost 1000. Instead, a line through all wines was calculated, which gives the rating that a wine of a particular price is expected to have. This allows each wine to be compared to other wines of the same price.

2 Dataset

The dataset was sourced from Kaggle, "Wine Reviews" (zynicide) [1], and contains reviews from *Wine Enthusiast* magazine. The file used was `winemag-data-130k-v2.csv`, with 129 971 reviews organized into 14 columns. Most of these columns were filtered out, focusing only on the points (rating), price (US$), country, title and description. The script cleans the data itself, so this does not have to be done manually by the user. First, rows missing the price, points or country were excluded from the calculations. Then, duplicate reviews were removed with the help of the title and description. This left 111 537 wines, around 86 % of the data, to be processed further.

3 Method

The data analysis was done in Python 3 (for example in VS Code), with the packages `pandas`, `numpy` and `matplotlib`. Before beginning the analysis, it was decided that the price would be analysed as the logarithm of the price. The prices are very skewed, with a median of \$25 and a maximum of \$3 300, and every doubling of the price adds roughly the same number of points. Using $\log_{10}(\text{price})$ therefore makes the relation between price and rating linear.

First, the expected rating was calculated with a least-squares line through all wines. Using one line for all wines means that every country is compared with the same standard:

$$\text{expected points} = a + b\,\log_{10}(\text{price}), \qquad a = 78.78, \quad b = 6.68. \tag{1}$$

Next, the value score of each wine was found by subtracting the expected points from the actual points:

$$\text{value score} = \text{actual points} - \text{expected points}. \tag{2}$$

A positive value indicates that a wine is better than other wines at the same price, whereas a negative value indicates that the wine is overpriced. The Pearson correlation coefficient r was also calculated to determine how strongly the log price and the rating go together; r^2 gives the share of the differences in rating that is explained by price.

A plot of all wines with the expected points (`fig1a` in the README) becomes very crowded due to the large amount of data. Therefore, Figure 1b shows only 3 000 random wines, together with the average points and their standard deviation (SD) in different price ranges, calculated from all wines. Afterwards, the wines were grouped by country, and a mean value score and median price were computed per country (Figure 2). A country needed at least 200 wines to be included, which allowed an analysis of fifteen countries. Finally, the uncertainties were calculated through subsampling: the wines were split into ten random subsamples, all results were recalculated in each subsample, and the standard deviation of the ten values was taken as the uncertainty.

4 Results

The fit gives expected points = 78.8+6.68 $\log_{10}$(price): a ten times more expensive wine scores 6.68±0.09 points more, which means that a wine gains only about 2 points when its price doubles. The Pearson correlation, $r = 0.618 \pm 0.003$, shows that more expensive wines tend to score better, but with $r^2 = 0.38$ only 38 % of the differences in rating are explained by price. The rest could be explained by other factors, such as the winery, the taster or the grape variety.

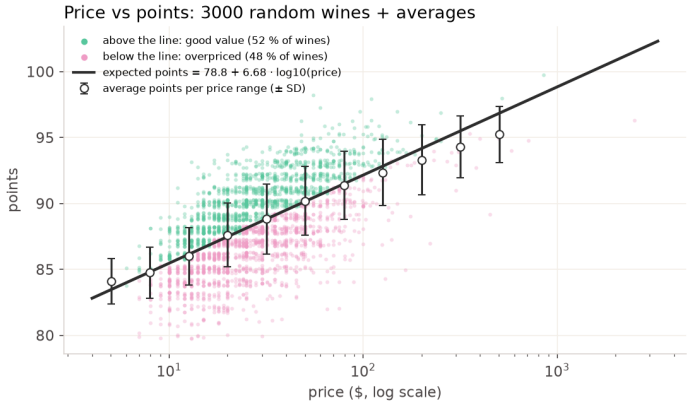


Figure 1b: Price versus rating: 3 000 random wines (green: above the line, pink: below), the fitted line giving the expected rating for each price, and the average rating per price range (± SD), calculated from all wines.

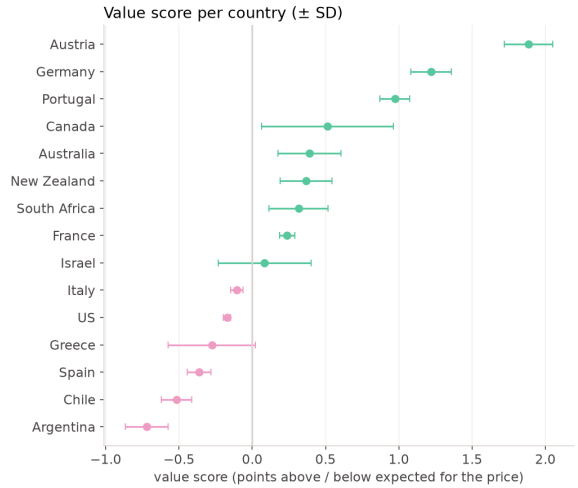


Figure 2: Mean value score per country (± SD of 10 random subsamples). Countries with a positive value score produce wines rated higher than their price predicts.

Figure 1b shows that the averages follow the line, so the model fits the data well. Wines at the same price differ by about ±3 points, and this spread determines whether a wine is good or bad value. Above around $200 the averages fall below the line, which shows that paying such prices is not compensated in quality.

Figure 2 answers the research question. Austria offers the best value, as its wines score on average 1.88 points above the line. This is the rating one would normally expect from wines about $10^{1.88/6.68} \approx 1.9$ times as expensive. Germany (+1.22) and Portugal (+0.97) follow, where Portugal combines good value with a low median price ($17). France and Italy are close to zero, meaning that one gets roughly what one pays for. Finally, Spain, Chile and Argentina offer cheap wines, but they lie below the line, meaning that they are still overpriced for their quality. This shows that cheap wines are not automatically good value, and neither are very expensive wines. Canada, Greece and Israel have larger uncertainties, as there are fewer wines to base the analysis on.

5 Limitations and conclusion

These results rely on a few assumptions. Wines from the same taster, winery or region are not fully independent, which means that the true uncertainties are probably somewhat larger than calculated. The ratings are also subjective, and only wines rated 80 or higher are published. Moreover, the prices are those of the US market at the time of review, so the results indicate which wines to buy rather than where to travel.

In conclusion, price clearly matters ($r = 0.62$), but it explains only 38 % of the differences in rating. The best value wine comes from Austria; however, when looking for wine that is both good and cheap, Portugal is the ideal choice.

References

[1] Z. Thoutt, *Wine Reviews*, Kaggle (2017). `https://www.kaggle.com/datasets/zynicide/wine-reviews`